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NITI Aayog

SECTOR AT A GLANCE 08

Electricity in India

Energy in India

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ELECTRICITY IN INDIA

ENERGY IN INDIA

The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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Overview

Over the past decade, India has made significant progress in increasing access to electricity. India has achieved nearly 99% household electrification, strengthening the distribution network and electricity access throughout the country.¹ From 2012 to 2022, per capita electricity consumption increased from 884 kWh to 1255 kWh, and total electricity demand increased from 8,24,301 GWh to 12,96,300 GWh, at a CAGR of 5%, making India the third-largest electricity consumer in the world.² The electricity consumption by the industry sector was highest at 41%, followed by the domestic and commercial sector at 34%, agriculture at about 18%, and transport and other applications consuming the rest.³ As of 2023, India had an installed capacity of 434 GW. Thermal power (coal, lignite, diesel and gas) comprised 56% of the total installed capacity.⁴ Renewable sources including solar, wind, biomass, small hydro, large hydro, and urban & industrial waste power account for 42%, and nuclear power accounts for 2% of total installed capacity.⁵

India has the potential to harness its abundant renewable energy resources for a smooth transition towards cleaner electricity generation, ensuring a sustainable and reliable power supply to meet the growing energy demands. Moreover, adopting decentralized renewable energy sources can empower consumers to generate electricity, especially in areas with limited electric grid connectivity.

Despite progress, currently electricity in India has high aggregate technical and commercial (AT&C) losses and poor financial condition of DISCOMs.

Given these opportunities and challenges, strengthening the financial situation of DISCOMs, reducing AT&C losses, and promoting generation, purchase, and consumption of renewable energy are the major policy focus areas. Several ongoing and recent central government initiatives, including but not limited to Ujjwal DISCOM Assurance Yojana, Green Energy Open Access Rules, Revamped Distribution Sector Scheme focus on expanding reliable and affordable access to clean electricity.

Advancing renewable energy integration through infrastructure enhancements and demand-side measures, accelerating existing efforts to improve DISCOMs' financial health, accelerating India's electric vehicle transition, and promoting energy storage innovation and deployment will be critical to continue advancing India's electricity sector. This note presents key trends related to different components of power in India and actions for the way forward.

Figure 1: Key electricity statistics

1617.82 TWh

Electricity output in India in 2022-23

6168.23 TWh

Projected electricity output in India in 2047

70% increase

In power generation capacity from 2014 – 2023

43%

Of India's total installed electricity capacity comes from non-fossil energy sources

1,57,000 HHs

targeted for last-mile connectivity and solar off-grid systems under the JANMAN scheme

Source: Ministry of Power, Growth of Electricity Sector in India, 2023; NITI Aayog, India Energy Security Scenario, accessed December 2023; Ministry of Power, Saubhagya Dashboard, accessed July 2023; Ministry of Information & Broadcasting, India's Power Sector Transformation: A Journey Towards Sustainable Energy and Universal Access, 2023; Press Information Bureau, Cabinet approves Pradhan Mantri Janjati Adivasi Nyaya Maha Abhiyan, 2023

1 Ministry of Power, Saubhagya Dashboard, accessed July 2023

2 Ministry of Statistics and Programme Implementation, Energy Statistics India, 2023; Central Electricity Authority, Growth of Electricity Sector in India from 1947-2020, 2020; Ministry of Power, Ministry of Power implements significant schemes to increase energy efficiency, 2022

3 Ministry of Statistics and Programme Implementation, Energy Statistics India, 2023

4 CEA, All India Installed Capacity (In MW) of Power Stations, 2023

5 CEA, All India Installed Capacity (In MW) of Power Stations, 2023

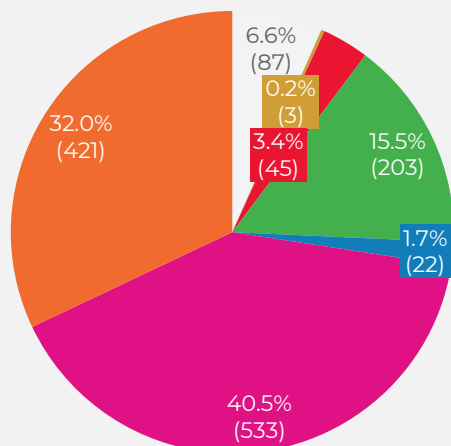
Figure 2: Electricity demand (including captive) and electricity supply (utility) in 2022 and 2047

Electricity demand (including captive), TWh

Miscellaneous Telecom Transport Buildings
Cooking Agriculture Industry

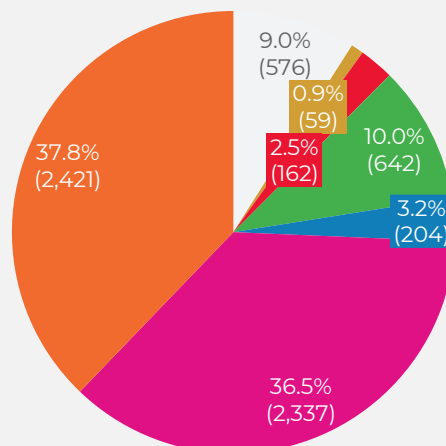
2022

Total demand is 1314.19 TWh



2047

Total demand is 6401.5 TWh

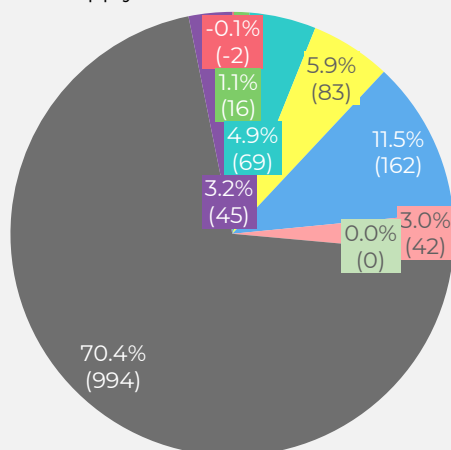


Electricity supply (utility), TWh

Electricity trade Wind Hydro CCS Gas
Bioenergy Solar Nuclear Coal

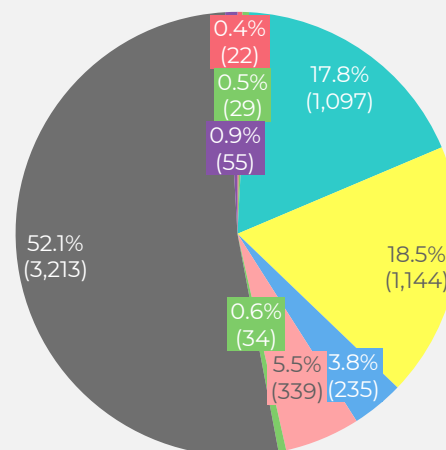
2022

Total supply is 1408.93 TWh



2047

Total demand is 6168.23 TWh



Source: NITI Aayog, India Energy Security Scenario, accessed December 2023

Opportunities and challenges

Opportunities

This section highlights the outcomes policymakers can achieve by transitioning to cleaner sources of electricity, leveraging decentralized energy solution, and adopting energy storage and green hydrogen.



Renewable energy expansion and decarbonization:

India aims to reduce the emission intensity of its GDP by 45% by 2030.⁶ Currently, thermal power plants, which produce over 50% of India's total electricity output, account for one-third of the total GHG emissions.⁷ As India strives to meet its decarbonization goals, the electricity sector has a significant opportunity to embrace a diversification strategy by transitioning to cleaner energy sources such as solar, wind, bio, hydro and atomic power.



Ensure access to reliable electricity:

With abundant potential for renewable energy sources like solar, wind, and hydro, India can pave a sustainable path to meet the surging electricity demand reliably. Investing in modern grid infrastructure, adopting smart grid solutions, and promoting energy storage technologies will bolster grid stability and reliability, ensuring uninterrupted power supply even in remote and underserved regions. The Government of India's latest time-of-day tariff rules enable the shifting of elastic electricity demand to solar hours, thus improving reliability.⁸ Further, since agricultural demand accounts for around 16% of the total electricity demand in 2022, a figure projected to remain at 10% in 2047, solarization in this sector has the potential to enhance both access to and the reliability of electricity supply for farmers.⁹



Storage solutions:

With India's renewable energy capacity growing rapidly, a tremendous opportunity for energy storage solutions has emerged.¹⁰ Policymakers expect to establish battery energy storage capacity of 27 GW by 2030 to meet the projected peak electricity demand of approximately 335 GW and an electrical energy requirement of 2280 BU.¹¹ The rapid expansion of renewable energy sources presents the challenge of effectively managing grid stability and mitigating the intermittency inherent in renewable sources like solar and wind. Energy storage technologies, including batteries and pumped hydro storage, have become indispensable in ensuring a reliable and resilient energy grid.



Opportunities for green hydrogen:

India's increasing renewable energy production can lower renewable energy costs, positively impacting the cost of green hydrogen production. As the cost of renewables decreases, producing green hydrogen through electrolysis powered by renewable sources becomes more competitive. This creates a unique opportunity for India to utilize its expanding renewable energy sector to promote the growth of green hydrogen.



Providing reliable power supply in remote areas through decentralized energy solutions:

⁶ Cabinet, India now stands committed to reduce Emissions Intensity of its GDP by 45 percent by 2030, 2022

⁷ ORF, Power Sector: Stumbling block in India's net-zero journey, 2022

⁸ Ministry of Power, Introduction of Time of Day Tariff, 2023

⁹ NITI Aayog, India Energy Security Scenario, accessed on December 2023

¹⁰ CEA, Central Electricity Authority, Report on optimal generation capacity mix for 2029-30, 2023

¹¹ CEA, Central Electricity Authority, Report on optimal generation capacity mix for 2029-30, 2023

Many remote regions in the country face challenges in accessing centralized power grids due to geographical barriers and infrastructural limitations. By adopting decentralized energy solutions such as off-grid solar, mini-grids, and small hydro projects, India can bring reliable electricity to areas with limited access to power resources.

Challenges

This section highlights some of the major challenges and constraints facing electricity in India. Efforts to resolve such challenges will ensure reliable, affordable, and clean electricity access to all.



Peak supply-demand electricity imbalance:

India's electricity sector faces a dual challenge of peak load deficit and insufficient power off-takers. The sector's complexity, rapid urbanization, and changing lifestyles contribute to peak load deficits during specific periods, leading to power outages. Simultaneously, the sector grapples with insufficient demand and power off-takers primarily due to financial distress in distribution companies.¹²



Renewable integration in the grid infrastructure:

The current grid infrastructure in several regions is not adequately equipped to integrate renewable energy at scale. Consequently, grid instability, curtailment of renewable power, and transmission losses have become prevalent issues. For instance, in Karnataka during 2019 and 2020, the state load dispatch centre (SLDC) had to reduce wind and solar generation by 10% to 25% in the middle of the day from June to September to ensure grid security. Research by the Center for Study of Science, Technology, and Policy indicates that curtailment of variable renewable energy could escalate further by 2030.¹³



Improving the financial health of DISCOMs:

AT&C losses in India's power transmission and distribution sector stood at 20% in 2022. This is notably higher than the global average of 7%. These losses adversely affect the performance and financial status of DISCOMs.¹⁴ Over the past twenty years, various efforts have been made to tackle the indebtedness of DISCOMs and enhance their financial condition, with the most recent being the Ujwal DISCOM Assurance Yojana (UDAY) scheme. These measures have shown progress in addressing the financial burdens of DISCOMs, assisting them temporarily. However, DISCOMs continue to encounter recurring losses.¹⁵ As of June 2023, the total dues of DISCOMs to suppliers (generation companies, transmission companies, and traders) were INR 44,386 Crore.¹⁶ As per the National Electricity Plan, India's capacity addition plans require INR 33,60,594 Crores between 2022 and 2032.¹⁷ Without improvement in DISCOMs' financial health, the investments required for new transmission and distribution networks, better integrate renewable energy, and reduce commercial losses will pose a challenge.¹⁸



Closing the investment gap in the power sector:

As per the National Electricity Plan, India's capacity addition plans require INR 33,60,594 Crores between 2022 and 2032.

12 CEEW, Offtake risk in renewable energy sector in India, 2019

13 International Energy Agency, Renewables Integration in India, 2021

14 International Energy Agency, India Energy Outlook, 2021; International Journal of Novel Research and Development, Losses in Transmission & Distribution Lines – a special reference to Gujarat Discom, 2022

15 PRS Legislative Research, Overview of the Power Sector, 2019

16 PIB, Remarkable improvement in recovery of outstanding dues of GENCOS and TRANSCOS: Shri R. K. Singh, Union Power & NRE Minister, 2023

17 Central Electricity Authority, National Electricity Plan (Volume I Generation), 2023

18 PRS Legislative Research, Overview of the Power Sector, 2019

Institutional framework and policy landscape —

Key ministries and departments

This section highlights the major central government bodies and national-level institutions that oversee electricity.



NITI Aayog

The NITI Aayog serves as the apex public policy think tank of the Government of India. Within the energy vertical, deals with five key sectors: power, coal, petroleum and natural gas, new and renewable energy, and atomic power. It conducts policy analysis on electricity generation, transmission and distribution; and undertakes steps to improve energy planning, and data management in India.



Ministry of Power

The ministry administers, formulates, and implements power sector-related policies and provides R&D and technical assistance relating to grid integration, transmission system networks and distribution systems in the States/UTs.



Ministry of New and Renewable Energy (MNRE)

The ministry develops renewable and clean energy technologies, systems, products, and services in line with global standards and deploys these for energy applications for sectors like power, transport and support country's objective of achieving energy security.



Bureau of Energy Efficiency

The bureau is the nodal authority for all GH2 projects in India, assists in developing policies and strategies to promote energy efficiency and conservation and utilises existing resources and infrastructure in performing functions under the Energy Conservation Act.



Indian Renewable Energy Development Agency

The agency finances and supports self-sustainable investments in energy generation from renewable sources, and carry out promotion of energy efficiency and sustainable environmental technologies



Department of Atomic Energy (DAE)

DAE is responsible for the entire spectrum of activities related to nuclear science and technology. This encompasses construction and operation of nuclear reactors, research and development of nuclear technologies, and exploration of atomic minerals.



Ministry of Coal

The ministry determines and implements policies and strategies related to the exploration and mining of coal and lignite reserves. It carries out activities like power generation, distribution, and marketing through entities under its administrative control.



Power Finance Corporation (PFC)

A public sector financial institution under the Ministry of Power, PFC finances the power sector through loans and investments. They play a crucial role in funding generation, transmission, and distribution projects, promoting sector growth.



REC Limited

A public sector financial institution under the Ministry of Power, REC primarily focuses on rural electrification and enhancing power infrastructure in remote areas. It finances and supports electrification projects, grid expansion, and distribution network improvement.



Solar Energy Corporation of India (SECI)

SECI, a government of India enterprise under MNRE, facilitates the development of India's solar energy market. They implement large-scale solar power projects, promoting grid-connected and off-grid solar power solutions.



Central Electricity Authority (CEA)

Functioning under the Ministry of Power, CEA acts as the apex technical and planning body for the power sector in India. They prepare the National Electricity Plan (NEP), a blueprint for power generation, transmission, and distribution, ensuring grid stability and meeting future electricity demands.



Ministry of Mines

The Ministry of Mines is responsible for surveying and exploring all minerals (except natural gases, petroleum, and atomic minerals). The ministry promotes responsible mining practices, mineral conservation, and sustainable development initiatives to support the country's energy and economic growth objectives.



Ministry of Petroleum and Natural Gas (MoPNG)

The central government ministry responsible for formulating and administering policies related to the exploration, development, production, refining, transportation, and marketing of petroleum and natural gas in India. MoPNG plays a key role in ensuring India's energy security through efficient management of hydrocarbon resources.



Ministry of Environment, Forests, and Climate Change (MOEFSEC)

MOEFSEC ensures environmental compliance in the power sector. It formulates policies related to environmental protection, forest conservation, and climate change mitigation. MOEFSEC's role is critical in balancing energy needs with ecological sustainability.

Major policies and programs

This section highlights existing policies related to integrated energy assessments and their most prominent features. A more comprehensive list of policies is provided in the policy repository of the NITI Knowledge Platform.



Electricity (Rights of Consumers) Amendment Rules, 2023:

Building on a 2020 amendment to ensure reliable and affordable supply of electricity, these latest rules introduce time-of-day tariffs that vary tariffs during solar hours and peak hours (where they are 10-20% higher), helping consumers manage consumption and supporting better grid integration of intermittent renewable energy. The latest amendment also simplifies smart metering rules, with no penal charges imposed on consumers based on maximum demand recorded prior to the smart meter's date of installation.¹⁹

¹⁹ PIB, Central Government Amends Electricity (Rights of Consumers) Rules, 2020 by Introducing Time of Day (ToD) Tariff and Simplification of Smart Metering rules, 2023

Electricity (Late Payment Surcharge and Related Matters) Rules, 2022:

The rules seek to enhance payment discipline in the power sector. They impose a late payment surcharge on payments delayed beyond a certain date, with the surcharge growing by 0.5% for each additional month of default (not exceeding 3% above the base rate). The rules also provide additional measures to limit non-compliance by establishing a payment security mechanism and progressive withdrawal of access. As of July 2023, the total outstanding dues of transmission companies and generation companies have halved since June 2022.²⁰

Revamped Distribution Sector Scheme (RDSS), 2021:

The RDSS aims to reduce AT&C losses to 12-15% across India and reduce the Average Cost of Supply to Average Revenue Realised (ACS-ARR) gap to zero by 2024-25. The Government of India has budgeted INR 3,03,758 Crores over a 5-year period (FY 2021-22 to FY2025-26).²¹ The Integrated Power Development Scheme (IPDS) and Deen Dayal Upadhyaya Gram Jyoti Yojana (DDUGJY) have been subsumed under RDSS. Under the scheme, financial assistance will be provided to DISCOMs for the upgradation of the distribution infrastructure based on meeting pre-qualifying criteria and achieving basic minimum benchmark in reforms.²² RDSS has also implemented a nationwide Smart Meter Program which envisages planned installation of 25 Crore smart meters across the country.²³

Ujwal DISCOM Assurance Yojana (UDAY), 2015:

The UDAY scheme is designed to achieve a financial turnaround of DISCOMs (Distribution Companies) in alignment with the government's goal of ensuring 24x7 affordable and accessible electricity for all. It aims to enhance operational efficiency, reduce power costs and interest expenses for DISCOMs, and enforce financial discipline through collaboration with state finances. States have the option to participate in the scheme by signing a tri-partite agreement with the Ministry of Power and their respective DISCOMs.²⁴

Pradhan Mantri Kisan Urja Suraksha evam Utthaan Mahabhiyan (PM KUSUM), 2019²⁵

The PM KUSUM scheme aims to ensure energy security for farmers in India, along with fulfilling India's 2030 commitments of achieving 40% installed power capacity from non-fossil-fuel sources through the following key provisions:

- Component A provides for setting up 10 GW of grid-connected, decentralised RE power plants. These plants (500 kW to 2 MW capacity) can be set up by individual farmers, groups, and other specified entities, wherein DISCOMs would purchase generated power at pre-determined tariffs
- Component B provides for the installation of 17.5 Lakh stand-alone solar agriculture pumps. Support will be provided to farmers for replacing existing diesel pumps with stand-alone solar water pumps (SWP) up to 7.5 HP in off-grid areas

20 PIB, Remarkable improvement in recovery of outstanding dues of GENCOS and TRANSCOS: Shri R. K. Singh, Union Power & NRE Minister, 2023

21 Ministry of Power, Government of India launches Revamped Distribution Sector Scheme (RDSS) to reduce the Aggregate Technical & Commercial (AT&C) losses to pan-India levels, 2023

22 Ministry of Power, Government of India launches Revamped Distribution Sector Scheme (RDSS) to reduce the Aggregate Technical & Commercial (AT&C) losses to pan-India levels, 2023

23 Ministry of Power, SMART METER NATIONAL PROGRAMME - A Model Standard Bidding Documents (SBD) for implementation of Smart Metering projects released for adoption by the States/Discoms – Union Power & NRE Minister Shri R. K. Singh, 2023

24 Government of India, UDAY for financial turnaround of Power Distribution Companies, 2015

25 National portal of India, Pradhan Mantri Kisan Urja Suraksha Evam Utthaan Mahabhiyan (PM KUSUM) scheme, accessed July 2023

- Component C provides for solarisation of 10 Lakh grid-connected agriculture pumps. This will support individual farmers to solarise their pumps, and DISCOMS will purchase excess power at pre-fixed tariffs



Green Energy Corridor Scheme²⁶

The Green Energy Corridor scheme aims to facilitate the evacuation and integration of renewable energy. It involves the developing transmission systems and control infrastructure to increase India's overall RE capacity by harnessing the potential of RE rich states like Andhra Pradesh, Gujarat, Himachal Pradesh, Jammu and Kashmir, Karnataka, Maharashtra, Rajasthan, Madhya Pradesh, and Tamil Nadu.



Green Energy Open Access Rules, 2022:

The rules promote the generation, purchase, and consumption of renewable energy for consumers (including commercial and industrial customers, small enterprises, and large households) to procure green energy, with DISCOMs being obligated to supply it to eligible customers (who in turn receive Green Certificates). The rules clarify the approval process for being granted open access, the charges and surcharges that will be levied for open access consumers, the renewable purchase obligations for DISCOMs, and the appropriate regulators and nodal agencies for approvals.²⁷ The launch of new market segments, such as Green Term Ahead Market (GTAM) and Green Day Ahead Market (G-DAM), facilitates the procurement of green energy through power exchanges. This procurement is then accounted for in the Renewable Purchase Obligation of obligated entities, thereby contributing to the deepening of the green electricity market.²⁸



Financial incentives for power sector reform:

The Ministry of Finance has been supporting states in the power sector by offering financial incentives through additional borrowing permissions. States can access an extra borrowing space of up to 0.5% of their GSDP annually from 2021-22 to 2024-25. In the past two years, states have raised INR 66,413 Crore, with INR 1,43,332 Crore available in 2023-24 for further reforms. The goal is to enhance operational efficiency, boost economic performance, and stimulate higher electricity consumption.²⁹

²⁶ Ministry of Power, Transmission Works under Green Energy Corridors-I, Accessed December 2023

²⁷ Ministry of Power, Ministry of Power notifies 'Green Energy Open Access' Rules to accelerate ambitious renewable energy programmes, 2022

²⁸ Ministry of Power, Ministry of Power formulates several steps to deepen the market for green electricity and to provide competitive price signals, 2023

²⁹ Ministry of Finance, Centre Provides Financial Incentives to States to accelerate Power Sector Reforms, 2023

Way forward

This section highlights considerations for policymakers seeking to advance policy formulation and implementation in this sector. These are based on reports and materials prepared by key government bodies and expert institutions like the Ministry of Power and the NITI Aayog.

 Enhancing renewable energy (RE) integration and upgrading infrastructure	- Focus on grid stability, modernization, and capacity enhancement to accommodate the variable nature of renewable energy and increased energy demand - Promote demand side flexibility and management to align consumption patterns with renewable energy generation, including through demand-side response measures that enable virtual power plants, time-of-day pricing, and smart metering - Provide policy support for storage solutions to balance supply and demand, and invest in smart grid technologies to improve the efficiency and reliability of energy supply - Encourage innovations for advanced storage technologies to improve the efficiency and cost-effectiveness of renewable energy storage
 Continue to improve and revive the financial health of DISCOMs	- Implement financial restructuring and debt relief measures to alleviate the financial burden on discoms - Pursue tariff rationalization and implement smart metering and billing to ensure cost recovery - Improve revenue realization and reduce losses through measures such as theft detection and technical loss reduction - Enhance coordination between central and state actors to ensure policy coherence and effective implementation
 Accelerate transition to EVs	- Expedite EV charging infrastructure coverage to facilitate the adoption and use of EVs across urban and rural areas (e.g., through greater deployment of public charging stations and support for home-based charging solutions) - Promote EV adoption through incentives and awareness campaigns, and enhance existing skilling programs to develop a strong skilled EV workforce - Provide regulatory support and financial incentives (e.g., many states have implemented investment and land related subsidies and electricity and stamp duty waivers) to develop affordable and efficient EV models (including for commercial use) and related infrastructure and storage solutions
 Innovating and deploying energy storage solutions	- Support research and development in advanced energy storage technologies (e.g., advanced chemistry cells) through the National Mission on Transformative Mobility and Battery Storage and complementary incentives from state governments - Facilitate the commercialization and deployment of new storage solutions - Create a conducive policy environment for the adoption and integration of energy storage technologies crucial for managing the variability of RE sources

Recommended readings

Ministry of Power. (2023). *Optimal Generation Capacity Mix for 2029-30*.

https://cea.nic.in/wp-content/uploads/irp/2023/05/Optimal_mix_report_2029_30_Version_2.0_For_Uploading.pdf

Ministry of Science and Technology. (2023). *Transforming the Indian Power Sector*.

<https://dst.gov.in/sites/default/files/DSO-White-Paper.pdf>

Ministry of Power. (2022). *Reforms in Power Sector*.

https://powermin.gov.in/sites/default/files/uploads/compilation_of_reforms.pdf

NITI Aayog and IEA. (2021). *Renewables Integration in India*.

[Renewables Integration in India – Analysis - IEA](#)

NITI Aayog and RMI. (2021). *Turning Around the Power Distribution Sector*.

https://www.niti.gov.in/sites/default/files/2021-08/Electricity-Distribution-Report_030821.pdf



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